

Evaluating ARP Spoofing Vulnerabilities in Modern Home and Enterprise Routers

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Abstract

This study investigates the susceptibility of modern home and enterprise routers to ARP spoofing attacks. We tested six routers from five manufacturers using tools like Bettercap and Wireshark in a controlled environment. Our findings reveal that all routers were vulnerable to ARP spoofing out of the box, with built-in defenses either disabled by default or ineffective. While AP/Client Isolation proved effective in one router, its limitations, such as restricting local device communication, make it impractical for many scenarios. Heuristic-based defenses like rate limiting and static ARP tables were effective but not scalable. These results highlight the need for manufacturers to enable robust defenses by default and for users to adopt proactive security measures to mitigate ARP spoofing risks in modern networks..

1 Introduction

Address Resolution Protocol (ARP) spoofing, also known as ARP poisoning, is a type of cyberattack in which an attacker sends falsified ARP messages over a local network. This malicious activity associates the attacker's Media Access Control (MAC) address with the IP address of a legitimate device, such as a router or a computer. As a result, the attacker can intercept, modify, or disrupt communication between devices on the network. ARP spoofing is particularly dangerous because it exploits the inherent trust-based nature of the ARP protocol, which lacks authentication mechanisms to verify the legitimacy of ARP messages. This vulnerability allows

attackers to execute man-in-the-middle (MITM) attacks, session hijacking, and denial-of-service (DoS) attacks, making ARP spoofing a persistent threat to network security.

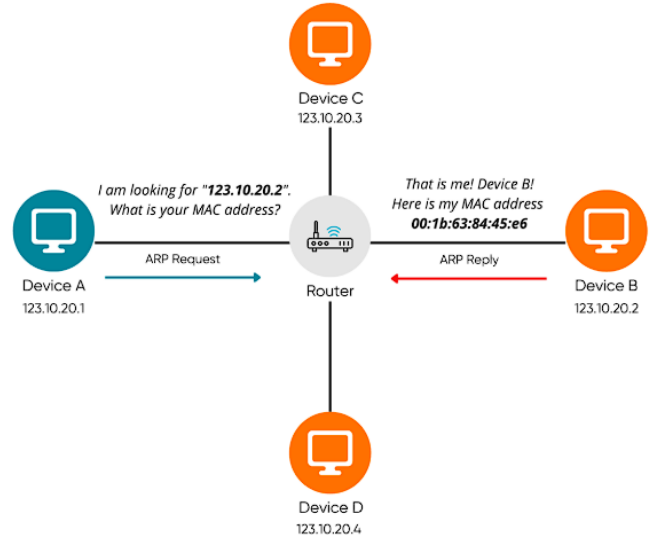


Figure 1: Address Resolution Protocol (ARP)

The ARP protocol, defined in 1982 by RFC 826 [3], was designed to map IP addresses to MAC addresses in local networks. Its simplicity and efficiency made it a cornerstone of network communication. However, this simplicity came at a cost: the protocol lacks any form of authentication, making it inherently vulnerable to spoofing attacks. The first documented ARP spoofing attacks emerged in the late 1990s, coinciding with the rise of the internet and the increasing com-

plexity of networked systems. Early attacks were primarily academic, demonstrating the theoretical vulnerabilities of ARP. However, as networks grew in size and importance, ARP spoofing became a practical tool for malicious actors. By the early 2000s, ARP spoofing was being used in real-world attacks, including data interception, session hijacking, and network disruption.

Over the years, ARP spoofing has evolved alongside advancements in network technology. In the early 2000s, tools like Ettercap and Cain and Abel made ARP spoofing accessible to even novice attackers, leading to a surge in its use. These tools automated the process of sending falsified ARP messages, making it easy to launch attacks with minimal technical knowledge. As a result, ARP spoofing became a common tactic in both targeted attacks and broad-based network intrusions. Despite the development of mitigation techniques, ARP spoofing remains a significant threat due to the widespread reliance on the ARP protocol in modern networks.

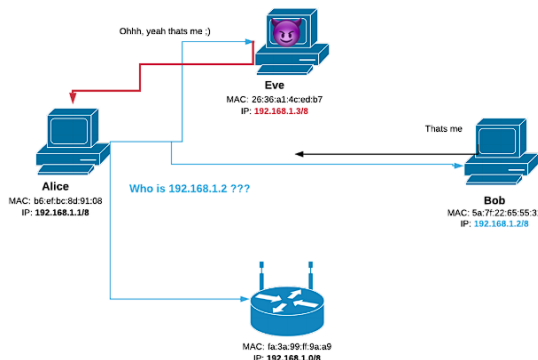


Figure 2: ARP Spoofing Attack Diagram

Several researchers and cybersecurity experts have contributed to the understanding and mitigation of ARP spoofing. Issac et al. [3] proposed a protocol called S-UARP by extending DHCP. It is a Centralized protocol. S-UARP request sent to DHCP+ only. It stores the IP-MAC mappings. Therefore, a reply takes place in an encrypted format with a secret key. Shukla et al. [8] proposed automatic configuration of

Static ARP entries. In this protocol, the static ARP entries are configured automatically with the help of the client-server protocol. It works in DHCP, static and MANET and large networks.

Despite these advancements, ARP spoofing remains a significant threat in modern networks. Home networks, in particular, are highly vulnerable due to the lack of advanced security features in consumer-grade routers. Enterprise networks, while better protected, are not immune. Many routers and switches include features like Dynamic ARP Inspection (DAI) and AP/Client Isolation, but these defenses are often disabled by default or require manual configuration. This leaves many networks exposed to ARP spoofing attacks, especially in environments where network administrators lack the expertise or resources to implement robust security measures.

The goal of this study is to evaluate the current state of ARP spoofing vulnerabilities in modern routers, both home and enterprise-grade. By testing six routers from five different manufacturers, we aim to answer the critical question: **Are modern routers still ARP spoofable?** Our findings will provide valuable insights into the effectiveness of built-in defenses and highlight the need for improved security measures in router design and configuration. This study is particularly timely given the increasing reliance on networked devices in both home and enterprise environments. As the Internet of Things (IoT) continues to grow, the potential impact of ARP spoofing attacks will only increase, making it essential to understand and mitigate these vulnerabilities.

The remainder of this report is organized as follows: Section 2 provides a detailed survey of related work, discussing existing research on ARP spoofing detection and mitigation techniques. Section 3 describes the methodology used in our experiments, including the tools and techniques employed to test ARP spoofing vulnerabilities. Section 4 presents the evaluation of our findings, highlighting the susceptibility of modern routers to ARP spoofing and the effectiveness of built-in defenses. Section 5 outlines future work, including recommendations for improving router security and engaging with manufacturers and the cybersecurity community. Finally, Section 6 lists the references used in this report.

2 Related Work

Address Resolution Protocol (ARP) spoofing, also known as ARP cache poisoning, has been a significant concern in network security for decades. Researchers have proposed various techniques to detect and mitigate ARP spoofing attacks, each with its own strengths and limitations. This section provides a detailed survey of the existing literature on ARP spoofing detection and mitigation techniques, categorizing them based on their methodologies and effectiveness.

2.1 Static ARP

One of the simplest and most effective methods to prevent ARP spoofing is the use of **Static ARP** entries. This technique involves manually mapping IP addresses to their corresponding MAC addresses in the ARP cache table. By doing so, only authorized devices with predefined mappings can communicate on the network. While this method is highly effective in small networks, it becomes impractical in large-scale environments due to the manual effort required to maintain the ARP table. Additionally, static ARP does not adapt well to dynamic networks where devices frequently join or leave the network [6].

2.2 Secure Unicast ARP (S-UARP)

Isaac et al. [3] proposed **Secure Unicast ARP (S-UARP)**, which extends the Dynamic Host Configuration Protocol (DHCP) to prevent ARP cache poisoning. In this approach, a DHCP+ server handles S-UARP packets, storing the IP-MAC mappings as it leases out IP addresses to hosts. Communication between hosts is initiated by sending an S-UARP request to the DHCP+ server, which replies with an encrypted response using a secret key. This method reduces broadcast congestion and prevents spoofed ARP replies. However, it requires modifications to the ARP protocol on all host devices and the deployment of a DHCP+ server, making it less feasible for widespread adoption.

2.3 ICMP-Based Detection

Jinhua et al. [4] proposed an **ARP spoofing detection algorithm** that uses the Internet Control Message Protocol (ICMP). This technique separates valid and invalid ARP packets, storing valid packets in a database. If invalid packets are detected, ICMP echo requests are injected to identify the attacker. While this method is effective against weak attackers, it has limitations, including a single point of failure and the need for backward compatibility. Additionally, it does not protect against strong attackers who can bypass ICMP-based detection.

2.4 Modified ICMP and Voting

Arote et al. [1] introduced a **modified ICMP and voting-based approach** to detect and prevent ARP spoofing. In this method, a central server monitors network traffic and uses ICMP ping packets to detect malicious hosts. A voting system is employed to elect a genuine central server, ensuring the integrity of the detection process. This technique is backward compatible and works well in centralized systems. However, it does not handle static entries that are not saved in the central server, limiting its effectiveness in certain scenarios.

2.5 Centralized Server-Based Detection

Srinath et al. [9] proposed a **centralized server-based approach** to detect and prevent ARP spoofing. In this method, a centralized server collects IP-MAC mappings from all hosts in the network and maintains a table of genuine hosts. If an attacker is detected, the destination host informs the centralized server, which then takes appropriate action. While this approach overcomes some limitations of voting-based methods, it does not perform detection during the registration process, allowing unauthenticated hosts to join the network.

2.6 Secure ARP (S-ARP)

Bruschi et al. [2] developed **Secure ARP (S-ARP)**, which uses public/private key cryptography to authenticate ARP replies. Each device in the network is assigned a key by an Authoritative Key Distributor (AKD), and all ARP replies are digitally signed by the sender. This method provides strong authentication and mitigates ARP spoofing attacks. However, it requires modifications to the ARP protocol on every host and the deployment of an AKD, making it less practical for large-scale networks.

2.7 Automatic Configuration of Static ARP Entries

Shukla et al. [8] proposed an **automatic configuration method** for static ARP entries. This approach uses a client-server protocol to automatically configure static ARP entries, eliminating the need for manual configuration. It is scalable and works well in Dynamic Host Configuration Protocol (DHCP), static, and Mobile Ad Hoc Network (MANET) environments. This method addresses the scalability and cost issues associated with traditional static ARP but requires additional infrastructure to support the automatic configuration process.

2.8 Software-Defined Networking (SDN) Based Methods

Masoud et al. [5] introduced an **SDN-based method** to detect and prevent ARP spoofing. This approach uses an SDN-enabled switch and a Ryu controller to monitor network traffic and detect ARP spoofing attacks. It supports both dynamic (SDN_DYN) and static (SDN_STA) IP addressing scenarios. While this method is effective in preventing ARP spoofing, it lacks intelligence and requires complex static configurations in large-scale SDN environments, which can degrade network performance.

2.9 Machine Learning-Based Detection

Sebbar et al. [7] utilized **machine learning techniques** to detect and mitigate Man-in-the-Middle (MITM) attacks in large-scale SDN environments. They developed a **Context-Based Node Acceptance-Random Forest (CBNA-RF)** model, which automatically detects malicious nodes and enforces security policies. This approach does not compromise network performance and provides automated defense mechanisms. However, it requires significant computational resources and training data to achieve high accuracy.

3 Methodology

In this study, we conducted a series of experiments to evaluate the susceptibility of modern home and enterprise routers to ARP spoofing attacks. The methodology involved setting up a controlled network environment, selecting a diverse set of routers, and using specialized tools to perform ARP spoofing attacks. This section outlines the experimental setup, tools used, and the step-by-step procedure followed during the experiments.

3.1 Experimental Setup

To ensure accurate and reproducible results, we created a controlled network environment. The setup included the following components:

- **Routers:** We tested six routers from five different manufacturers, including one enterprise-grade router and five home routers. The routers selected for the experiment were:
 - Netgear Nighthawk X6 R8000 (June 2014)
 - Tenda RX2 Pro (November 2022)
 - DLink DIR-3040 (May 2020)
 - ASUS RT AX-57 (September 2021)
 - TP-Link EAP613 (February 2023) – Two units of this enterprise-grade router were tested.

- **Software Tools:** The following tools were used to perform ARP spoofing attacks and monitor network traffic:
 - **Bettercap:** A powerful network attack tool used to perform ARP spoofing and monitor network traffic.
 - **Wireshark:** A network protocol analyzer used to capture and analyze ARP packets during the experiments.
- **Environment:** The experiments were conducted using Kali Linux (Version 2024.4) running on a live USB for better stability and direct hardware access. Initially, we attempted to use Kali Linux in a virtual machine (VM), but encountered performance and networking issues. Switching to a live USB resolved these issues.

3.2 Step-by-Step Procedure

The experiments were conducted in the following steps:

1. Network Configuration:

- Each router was set up in a controlled environment with a small local area network (LAN) consisting of three devices: the router, a victim device, and an attacker device.
- The victim device was connected to the router to simulate a legitimate user, while the attacker device was used to perform ARP spoofing attacks.

2. ARP Spoofing Attack:

- Using Bettercap, we initiated ARP spoofing attacks on each router. The attacker device sent falsified ARP messages to associate its MAC address with the IP address of the router.
- The attack was performed in two phases:
 - **Phase 1:** The attacker sent ARP replies to the victim device, poisoning its ARP cache and redirecting traffic to the attacker.

- **Phase 2:** The attacker monitored the traffic between the victim and the router using Wireshark to verify the success of the attack.

3. Observation and Analysis:

- For each router, we observed whether the router detected or prevented the ARP spoofing attack.
- If the router had built-in defenses (e.g., Dynamic ARP Inspection or AP/Client Isolation), we tested their effectiveness by enabling them and repeating the attack.
- We documented the results, including whether the attack was successful, whether the router detected the attack, and the effectiveness of any built-in defenses.

4. Testing Built-in Defenses:

- For routers with built-in defenses, we tested their effectiveness by enabling them and repeating the ARP spoofing attack.
- We also evaluated whether these defenses were enabled by default or required manual configuration.

3.3 Challenges Faced

During the experiments, we encountered several challenges:

- **Environment Setup Issues:** Initially, we used Kali Linux in a virtual machine (VM), but encountered performance and networking issues. Switching to a Kali Linux live USB resolved these issues.
- **Router Configuration Complexity:** Some routers had complex settings, making it difficult to locate or enable ARP spoofing defenses. We referred to user manuals and online forums for guidance.
- **False Positives/Negatives:** Some built-in defenses appeared to work initially but failed under closer scrutiny. We repeated tests multiple times to ensure accuracy.

4 Evaluation

In this section, we present the key findings from our experiments, including proof of ARP spoofing attacks and the effectiveness of built-in defenses in modern routers. We also provide visual evidence from our experiments to support our conclusions.

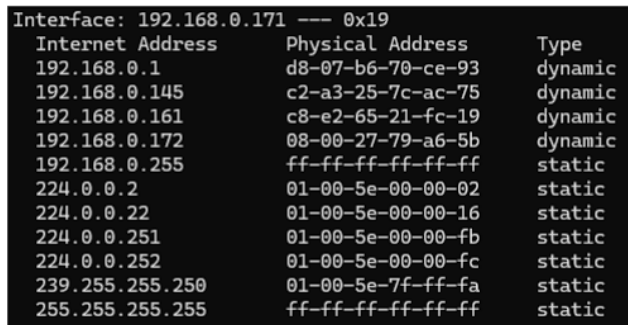
4.1 Key Findings

Our experiments revealed the following key findings:

- **All Routers Were ARP Spoofable:** Despite being modern devices, all six routers tested were vulnerable to ARP spoofing attacks out of the box. This includes both home and enterprise-grade routers.
- **Built-in Defenses Were Ineffective or Disabled by Default:** Some routers had built-in defenses such as Dynamic ARP Inspection (DAI) or AP/Client Isolation. However, these defenses were either disabled by default or proved ineffective in preventing ARP spoofing attacks.
- **AP/Client Isolation Worked in One Router:** We found that enabling **AP/Client Isolation** in one router (ASUS RT AX-57) effectively prevented ARP spoofing. However, this defense has significant limitations, such as restricting local device communication, making it impractical for many use cases.
- **Heuristic-Based Defenses Were Partially Effective:** Techniques like rate limiting of ARP packets and static ARP tables were effective in some cases but were not scalable for larger networks.

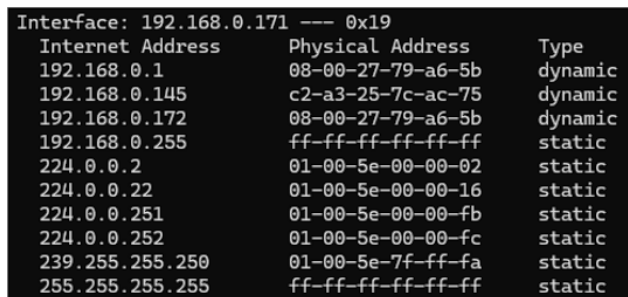
4.2 Proof of ARP Spoofing Attacks

To demonstrate the success of ARP spoofing attacks, we captured screenshots of the ARP cache before and after the attack. Below are the results for two routers:



Interface: 192.168.0.171 --- 0x19		
Internet Address	Physical Address	Type
192.168.0.1	d8-07-b6-70-ce-93	dynamic
192.168.0.145	c2-a3-25-7c-ac-75	dynamic
192.168.0.161	c8-e2-65-21-fc-19	dynamic
192.168.0.172	08-00-27-79-a6-5b	dynamic
192.168.0.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Figure 3: ARP Cache Before Spoofing on Tenda RX2 Pro



Interface: 192.168.0.171 --- 0x19		
Internet Address	Physical Address	Type
192.168.0.1	08-00-27-79-a6-5b	dynamic
192.168.0.145	c2-a3-25-7c-ac-75	dynamic
192.168.0.172	08-00-27-79-a6-5b	dynamic
192.168.0.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Figure 4: ARP Cache After Spoofing on Tenda RX2 Pro

4.2.1 Tenda RX2 Pro

In Figure 3, the ARP cache shows the legitimate MAC address of the router. After performing the ARP spoofing attack, the ARP cache was poisoned, as shown in Figure 4, where the attacker’s MAC address is associated with the router’s IP address.

4.2.2 DLink DIR-3040

Similarly, for the DLink DIR-3040, Figure ?? shows the legitimate ARP cache, while Figure 6 demonstrates the success of the ARP spoofing attack, with the attacker’s MAC address replacing the legitimate one.

```
C:\Users\ASUS>arp -a

Interface: 192.168.100.109 --- 0x4
Internet Address      Physical Address      Type
192.168.100.1         78-98-e8-32-8b-ab     dynamic
192.168.100.155       94-08-53-6c-b1-37     dynamic
192.168.100.255       ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Figure 5: ARP Cache Before Spoofing on DLink DIR-3040

```
C:\Users\ASUS>arp -a

Interface: 192.168.100.109 --- 0x4
Internet Address      Physical Address      Type
192.168.100.1         94-08-53-6c-b1-37     dynamic
192.168.100.155       94-08-53-6c-b1-37     dynamic
192.168.100.255       ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\ASUS>
```

Figure 6: ARP Cache After Spoofing on DLink DIR-3040

4.3 Available Defenses

We also evaluated the built-in defenses available in some routers. Below are the results:

4.3.1 AP/Client Isolation in ASUS RT AX-57

The ASUS RT AX-57 router featured a built-in defense called **AP/Client Isolation**. When enabled, this feature prevented devices on the same network from communicating with each other, effectively mitigating ARP spoofing attacks. However, this defense has significant limitations:

- Devices cannot communicate locally, which disrupts file sharing, printing, and other local network services.
- Smart devices that rely on local communication (e.g., IoT devices) may not function properly.

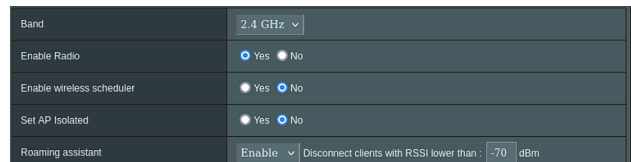


Figure 7: AP/Client Isolation Settings in ASUS RT AX-57

4.3.2 Rate Limiting and Static ARP Tables

Some routers supported heuristic-based defenses like rate limiting of ARP packets and static ARP tables. While these defenses were effective in small networks, they were not scalable for larger environments. Additionally, configuring static ARP tables manually is impractical for most users.

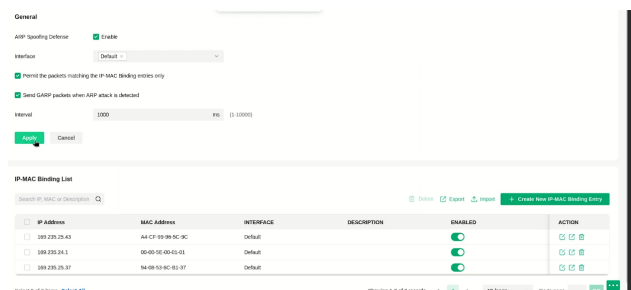


Figure 8: Rate Limiting and static ARP Table Settings in TP-Link EAP613

5 Future Work

Our experiments revealed significant vulnerabilities in modern routers, both home and enterprise-grade, despite the availability of built-in defenses. Based on these findings and feedback from our professor, we propose the following directions for future work:

5.1 Engaging with Router Manufacturers and Communities

One of the most critical steps is to raise awareness about the vulnerabilities and shortcomings of current

router defenses. We plan to:

- **Post Findings in Manufacturer Forums:** We will share our results on the official forums of router manufacturers (e.g., Netgear, ASUS, TP-Link, DLink, Tenda) to highlight the issues with their devices. This includes:
 - The fact that built-in defenses are often disabled by default.
 - The ineffectiveness of some defenses even when enabled.
 - The need for more robust and user-friendly security features.
- **Engage with the Cybersecurity Community:** We will post our findings on cybersecurity forums and communities (e.g., Reddit’s r/netsec, Stack Exchange, and specialized security blogs) to encourage discussion and collaboration on improving router security.

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